CMPG 325 — Technical Report

Design, Implementation and Testing

Student: Mthimunye, Sakhile (45224706)

Project ID: CMPG325-2026-095

Client ID: CLI-095

Client: Labour Department Taung Office (Government)

Challenge: Default Routing (edge/ISP path design)

Department of Computer Science and Information Systems

North-West University

Contents

1 Introduction **4**

1.1 Assigned parameters 4

1.2 Project timeline 4

2 Client Requirements Analysis **4**

2.1 Organisational context 4

2.2 Functional requirements 5

2.3 Non-functional requirements 5

2.4 Interpreting the constraint and the change request 5

3 Network Design **5**

3.1 Physical topology 5

3.2 Logical topology 7

3.3 Design decisions 7

4 IP Addressing Plan **8**

4.1 Method 8

4.2 Addressing table 8

4.3 Static and dynamic addresses 8

4.4 How CR5 is absorbed without renumbering 9

5 Implementation in Packet Tracer **9**

5.1 Build order 9

5.2 Switching 10

5.3 Routing, DHCP relay and NAT on R-EDGE 10

5.4 Simulated ISPs and internet host 11

6 Assigned Challenge: Default Routing (Edge/ISP Path Design) **12**

6.1 What was configured 12

6.2 Why this is appropriate 13

6.3 How the Finance constraint is met 13

6.4 Verification 13

6.5 NAT during failover 15

7 Additional Feature: Reception Access Restriction **15**

8 Testing Summary **18**

9 Troubleshooting **18**

10 GitHub Portfolio and Video **19**

10.1 Portfolio 19

10.2 Video demonstration 19

11 Reflection and Future Improvements **19**

11.1 What was learned 19

11.2 Limitations 19

11.3 Possible improvements 20

12 Conclusion **20**

A Device Configurations **21**

A.1 R-EDGE 21

A.2 SW-CORE 22

A.3 SW-FIN (example access switch) 23

A.4 Simulated ISP devices 24

List of Figures

1 Physical topology: dual ISP links, edge router, core switch, five department access switches. 6

2 Logical topology: VLANs and subnets, the two default routes, and the Reception access control. 7

3 `show vlan brief` on SW-CORE: all six VLANs are active. 10

4 `show interfaces trunk` on SW-CORE: six trunks, native VLAN 99, and VLANs 10, 20, 30, 40, 50 and 99 allowed on each. 10

5 `show ip route` on R-EDGE with the primary path active: every VLAN subnet is directly connected and the gateway of last resort is `192.168.41.146`. 11

6 A PC in Employment Services pinging `8.8.8.8`: four replies, no loss. 12

7 `show ip nat translations` on R-EDGE: private source `192.168.41.35` translated to `192.168.41.145`, the primary WAN interface address. 12

8 Before the outage: gateway of last resort `192.168.41.146` (16 subnets in the table). 13

9 After `shutdown` on `Gi0/0/1`: gateway of last resort `192.168.41.150` (14 subnets). 14

10 A Finance PC pinging `8.8.8.8` over the backup path: four replies, no loss. . . . 14

11 After `no shutdown` on `Gi0/0/1`: gateway of last resort back to `192.168.41.146` (16 subnets). 15

12 A Reception PC pinging Finance hosts: “destination host unreachable” from the Reception gateway, 100% loss. 16

13 A Management PC pinging Finance hosts: three of four replies each time; the first request of each run times out. 17

14 A Reception PC pinging the intranet name: the name resolves to 192.168.41.85, while ICMP to that subnet is refused. 17

1 Introduction

This report documents the design, implementation, testing and evidence for a simulated network built for the Labour Department Taung Office, using Cisco Packet Tracer. It covers the whole project from the client brief to the finished, tested network, and it is accompanied by the Packet Tracer file, the GitHub portfolio of evidence and the individual video demonstration.

1.1 Assigned parameters

The parameters below come directly from the project brief and were used without change.

Item	Detail
Client	Labour Department Taung Office (Taung), Government
Addressing block	`192.168.41.0/24`
Assigned networking challenge	Default Routing (edge/ISP path design), Intermediate
Design constraint	A backup internet path is required for the Finance function
Change request CR5	User numbers grow by 25%; the addressing plan must absorb this without renumbering

1.2 Project timeline

Stage	Date	Content
Project commencement	14 August 2026	Client brief received
Milestone 1	28 August 2026	Client requirements, topology, IP addressing plan, initial GitHub repository
Milestone 2	2 October 2026	Working Packet Tracer file, assigned feature, testing evidence, updated portfolio
Final submission	16 October 2026	Packet Tracer file, GitHub portfolio, technical documentation, individual video

2 Client Requirements Analysis

2.1 Organisational context

The Taung Labour Department office is a small government branch office. The brief gives the organisation and industry but no staff numbers or department list, so the departmental structure and the headcounts used for sizing are **planning assumptions** based on what a Labour Department branch typically does. They are stated explicitly wherever they affect a design decision, so they can be adjusted without changing the method.

The network serves five functional groups:

- **Management**: office management and administration.
- **Employment Services**: case workers handling job-seeker registration and claims; the largest group.
- **Finance**: payroll and procurement; the group named in the design constraint.

- **IT / Servers**: internal servers (DHCP and DNS) and network management.
- **Reception**: front desk and public-facing devices.

2.2 Functional requirements

- Logical separation of each department using VLANs.
- Automatic addressing (DHCP) and name resolution (DNS) from a central server.
- Inter-VLAN routing so departments can reach shared services and the internet.
- A redundant internet path, with a primary and a backup ISP link.
- Default routing configured at the network edge, as the assigned technical challenge.
- An addressing plan with built-in headroom so CR5 needs no renumbering.
- A working, testable Packet Tracer implementation with documented evidence.

2.3 Non-functional requirements

- **Security:** the public-facing Reception network must not be able to reach Finance or IT.
- **Reliability:** no single point of failure on the internet edge.
- **Scalability:** 25% user growth in any department without redesign.
- **Traceability:** every design decision and test is evidenced in the GitHub portfolio.

2.4 Interpreting the constraint and the change request

Design constraint. "Backup internet path required for the finance function" was interpreted as: Finance must keep internet access if the primary ISP link fails. All departments share one edge router, so the backup path is provided at the edge and Finance is protected together with everyone else. This satisfies the stated requirement with the simplest design that fits an Intermediate challenge. A Finance-only path would need policy-based routing or a second edge device, which the brief does not ask for.

Change request CR5. "User numbers grow by 25% — the addressing plan must absorb this without renumbering" was interpreted as: every subnet must already be large enough for its post-growth headcount, so extra users only consume addresses already inside the existing range. No VLAN, subnet, gateway or existing device address may change. Section 4 shows the numbers.

3 Network Design

3.1 Physical topology

Two ISP links feed the edge router R-EDGE. R-EDGE connects to the core switch SW-CORE, which trunks to five access switches, one per department. End devices (PCs, laptops and printers) connect to their department's access switch, and the central server Server0 connects to the IT switch.

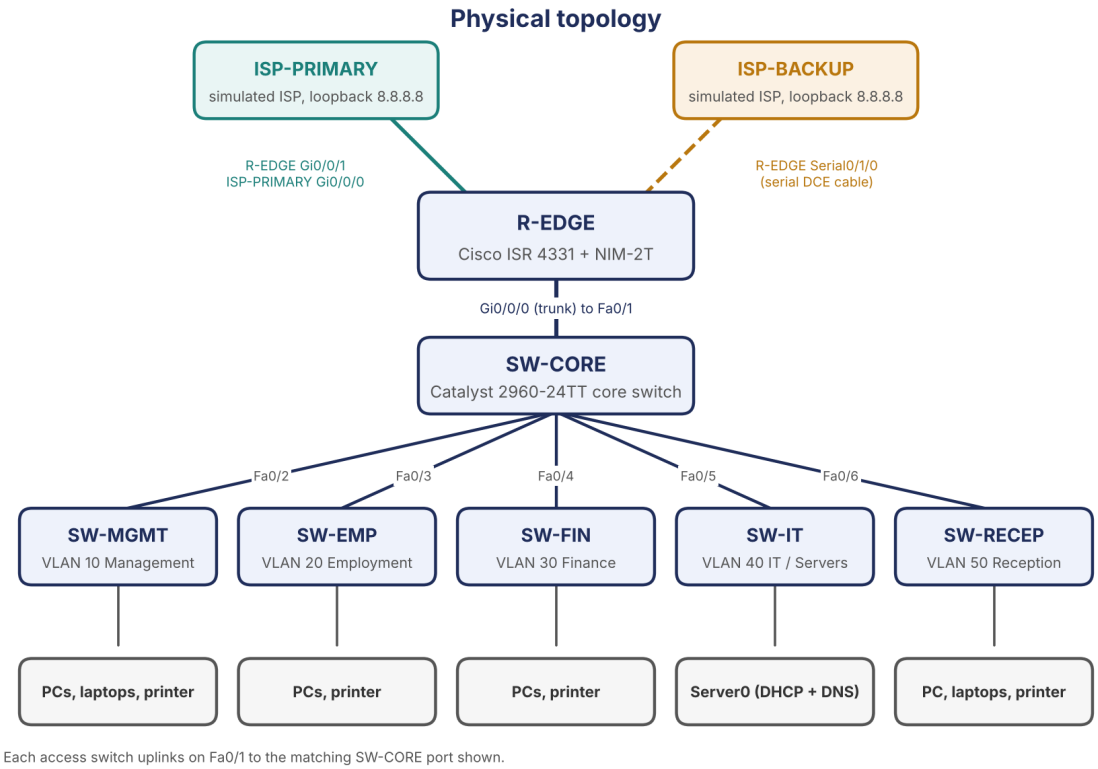


Figure 1: Physical topology: dual ISP links, edge router, core switch, five department access switches.

R-EDGE is a Cisco ISR 4331. It has two onboard Gigabit Ethernet ports, but the design needs three physical links (primary ISP, backup ISP, LAN trunk). The onboard `Gi0/0/0` carries the LAN trunk, `Gi0/0/1` connects to the primary ISP, and a **NIM-2T** serial module adds `Serial0/1/0` for the backup ISP. The reason for choosing a serial module is explained in Section 9.

Device	Model	Role
R-EDGE	ISR 4331 + NIM-2T	Edge router: WAN links, inter-VLAN routing, NAT, DHCP relay, ACL
SW-CORE	2960-24TT	Core switch: trunks to R-EDGE and all access switches
SW-MGMT, SW-EMP, SW-FIN, SW-IT, SW-RECEP	2960-24TT	Department access switches
Server0	Generic server	DHCP and DNS, in the IT VLAN
ISP-PRIMARY, ISP-BACKUP	ISR 4331	Simulated ISPs, each with a loopback acting as an internet host

3.2 Logical topology

R-EDGE performs inter-VLAN routing with one 802.1Q sub-interface per VLAN on `Gi0/0/0` (router-on-a-stick). A primary default route points to ISP-PRIMARY and a floating static default route points to ISP-BACKUP.

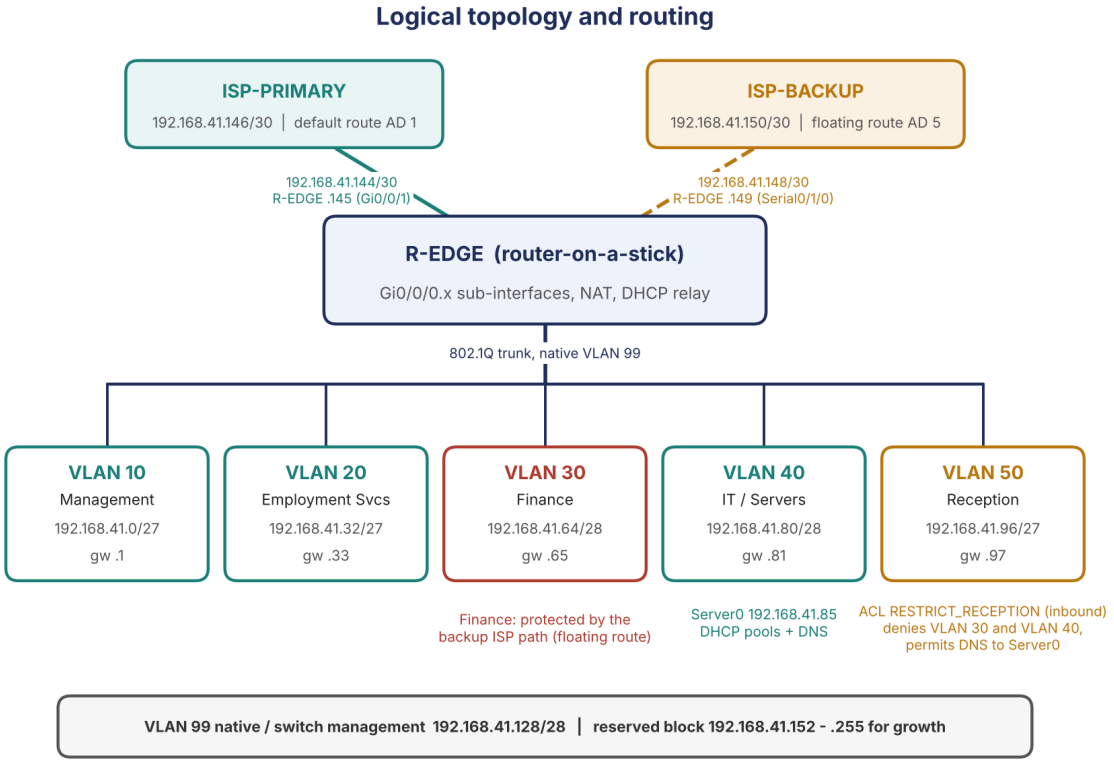


Figure 2: Logical topology: VLANs and subnets, the two default routes, and the Reception access control.

3.3 Design decisions

Decision	Justification
Router-on-a-stick inter-VLAN routing	One edge router keeps the design scoped to an Intermediate challenge. A separate layer-3 core switch is not required by the brief.
Floating static default route for the backup ISP	Directly implements the assigned challenge and meets the backup-path constraint without a dynamic routing protocol.
One VLAN per department	Contains broadcast traffic and allows department-level access control.
Extended ACL on the Reception sub-interface	Reception is public-facing, while Finance and IT hold sensitive functions. DNS to Server0 is permitted first so name resolution still works.
DHCP and DNS in the IT VLAN	Central services, reachable from every VLAN through inter-VLAN routing and DHCP relay.
Headroom built into every subnet	Absorbs CR5 without renumbering.
Other departments left open to each other	The brief does not require broader segmentation, and extra rules would add complexity without serving a requirement.

4 IP Addressing Plan

4.1 Method

The block `192.168.41.0/24` (256 addresses) was divided with variable-length subnet masking (VLSM). A single equal split would waste addresses, because Employment Services needs room for about 25 hosts while Finance needs about 10. Each subnet was therefore sized to the smallest block that holds its post-growth headcount:

Mask	Block size	Usable hosts	Used for
/27	32	30	Management, Employment Services, Reception
/28	16	14	Finance, IT/Servers, switch management
/30	4	2	The two router-to-ISP links

Every block starts on a multiple of its own size, so the subnets do not overlap. They are placed one after another with no gaps up to `192.168.41.151`, and the remainder is kept as a reserve.

4.2 Addressing table

VLAN	Segment	Subnet	Usable range	Gateway	Broadcast	Hosts
10	Management	192.168.41.0/27	.1 – .30	.1	.31	30
20	Employment Services	192.168.41.32/27	.33 – .62	.33	.63	30
30	Finance	192.168.41.64/28	.65 – .78	.65	.79	14
40	IT / Servers	192.168.41.80/28	.81 – .94	.81	.95	14
50	Reception	192.168.41.96/27	.97 – .126	.97	.127	30
99	Switch management	192.168.41.128/28	.129 – .142	.129	.143	14
–	R-EDGE / ISP-PRIMARY	192.168.41.144/30	.145 – .146	–	.147	2
–	R-EDGE / ISP-BACKUP	192.168.41.148/30	.149 – .150	–	.151	2
–	Reserved	192.168.41.152 – .255	–	–	–	104

The WAN link subnets are taken from the assigned block for the simulation. In a real deployment the ISP would normally assign them.

4.3 Static and dynamic addresses

- **Router gateways:** the first usable address of each department subnet; VLAN 99 uses `.129`.
- **Switch management:** SW-CORE `.130`, SW-MGMT `.131`, SW-EMP `.132`, SW-FIN `.133`, SW-IT `.134`, SW-RECEP `.135`, all in VLAN 99.
- **Server0:** static `192.168.41.85/28`, gateway `.81`, in the IT VLAN.
- **DHCP pools** on Server0 for the four user VLANs, each starting just above the gateway and stopping below the last usable address:

Pool	Gateway	Lease range	Left for static devices
Management	192.168.41.1	.2 – .29	.30
Employment Services	192.168.41.33	.34 – .61	.62
Finance	192.168.41.65	.66 – .77	.78
Reception	192.168.41.97	.98 – .125	.126

Printers are given static addresses in the space above each pool so they can never collide with a DHCP lease. The IT VLAN has no pool because its devices are static. All pools use DNS server `192.168.41.85`.

4.4 How CR5 is absorbed without renumbering

Each subnet was sized against its post-growth headcount. The current headcounts below are planning assumptions, since the brief does not give any.

Department	Assumed users	After +25%	Usable hosts	Spare after growth
Management	5	7	30	23
Employment Services	20	25	30	5
Finance	8	10	14	4
IT / Servers	5	6	14	8
Reception	10	13	30	17

After growth every department still fits inside the subnet it already has. The extra users take addresses that were free from the first day, so no VLAN, subnet, gateway or existing device address changes. Employment Services has the least spare room (5 addresses), so the unallocated block `192.168.41.152` to `.255` is kept as a documented reserve for any further growth.

5 Implementation in Packet Tracer

5.1 Build order

Step	Action
1	Place and name all devices; fit the NIM-2T module to R-EDGE before cabling.
2	Cable the network (copper straight-through; serial DCE for the backup ISP).
3	Configure SW-CORE, then each access switch: VLANs, trunks, access ports, management address.
4	Configure R-EDGE: sub-interfaces, WAN links, default routes, NAT, ACL.
5	Configure the two simulated ISP devices.
6	Configure Server0 with DHCP pools and a DNS record.
7	Set end devices to DHCP and test.

Complete configurations for every device are in the `configs` folder of the repository and in Appendix A.

5.2 Switching

VLANs 10, 20, 30, 40, 50 and 99 were created on SW-CORE and on each access switch. SW-CORE has an 802.1Q trunk to R-EDGE on `Fa0/1` and to each access switch on `Fa0/2` to `Fa0/6`. All trunks use native VLAN 99 and allow only the VLANs in use. Access ports are assigned to their department VLAN with PortFast enabled.

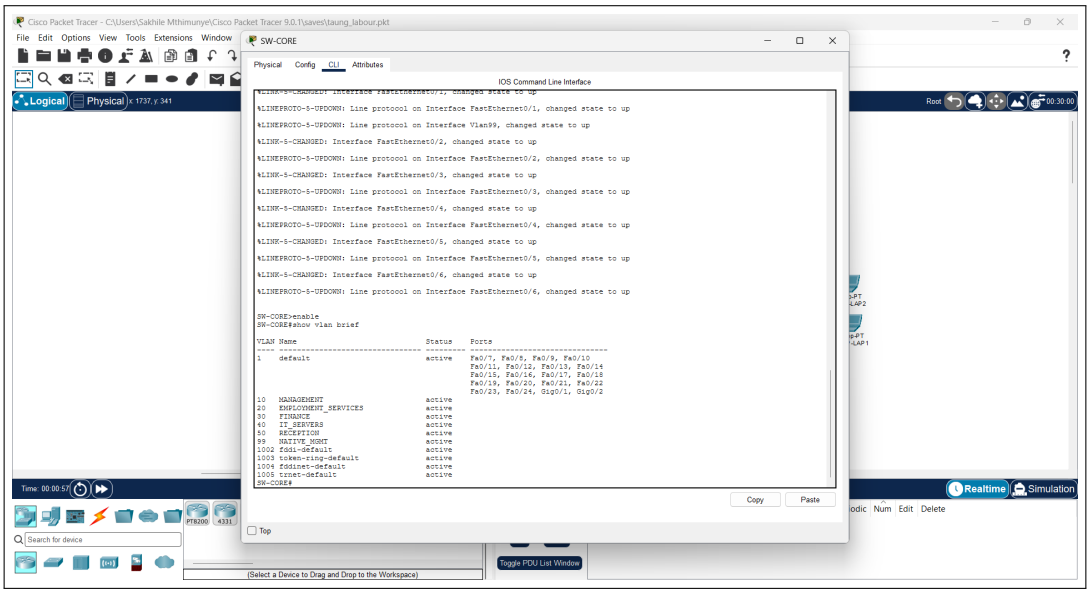


Figure 3: `show vlan brief` on SW-CORE: all six VLANs are active.

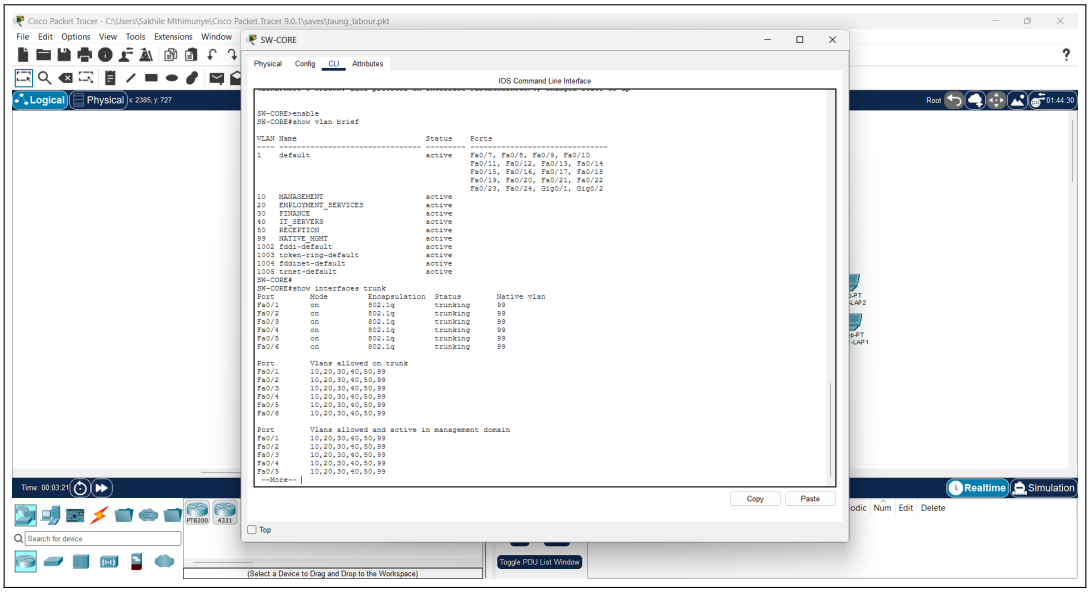


Figure 4: `show interfaces trunk` on SW-CORE: six trunks, native VLAN 99, and VLANs 10, 20, 30, 40, 50 and 99 allowed on each.

5.3 Routing, DHCP relay and NAT on R-EDGE

R-EDGE has one sub-interface per VLAN on `Gi0/0/0`. Each user VLAN also has `ip helper-address 192.168.41.85` so DHCP requests reach Server0 across VLANs. The Finance sub-interface shows the pattern:

```
interface GigabitEthernet0/0/0.30
```

```
description VLAN 30 - Finance
encapsulation dot1Q 30
ip address 192.168.41.65 255.255.255.240
ip helper-address 192.168.41.85
ip nat inside
```

The two WAN interfaces are marked `ip nat outside`. Private addresses are translated with overload (PAT) on the active outside interface, using the standard access list `NAT_ACL` that permits `192.168.41.0/24`.

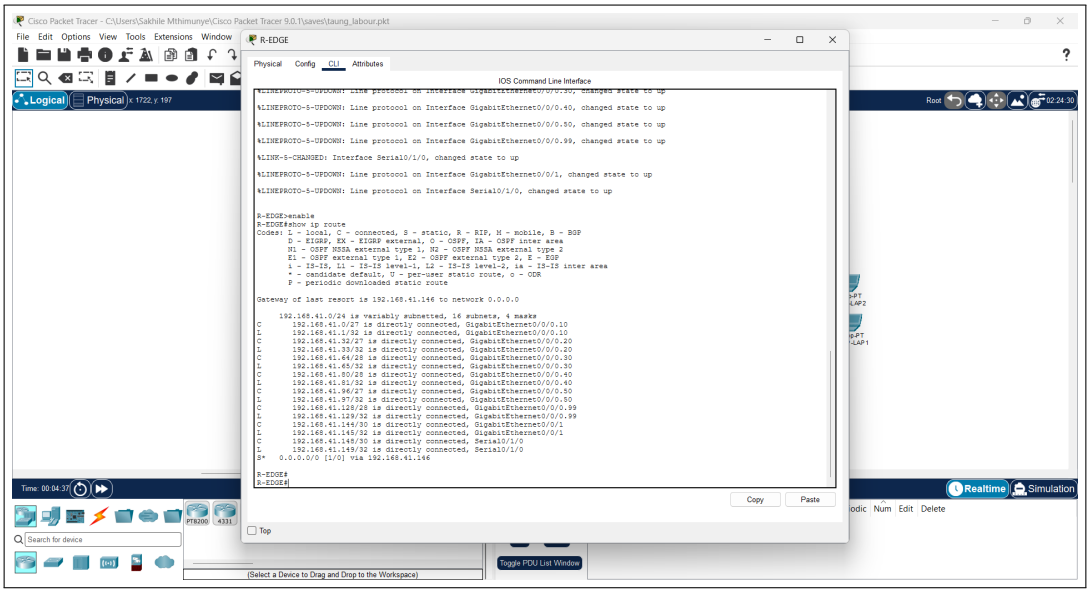


Figure 5: `show ip route` on R-EDGE with the primary path active: every VLAN subnet is directly connected and the gateway of last resort is `192.168.41.146`.

5.4 Simulated ISPs and internet host

Packet Tracer has no internet, so each ISP device carries a loopback interface `8.8.8.8/32` that acts as a stand-in internet host, plus a static route back to `192.168.41.0/24` through R-EDGE so replies can return. Without these, a ping to `8.8.8.8` has nothing to answer it (see Section 9).

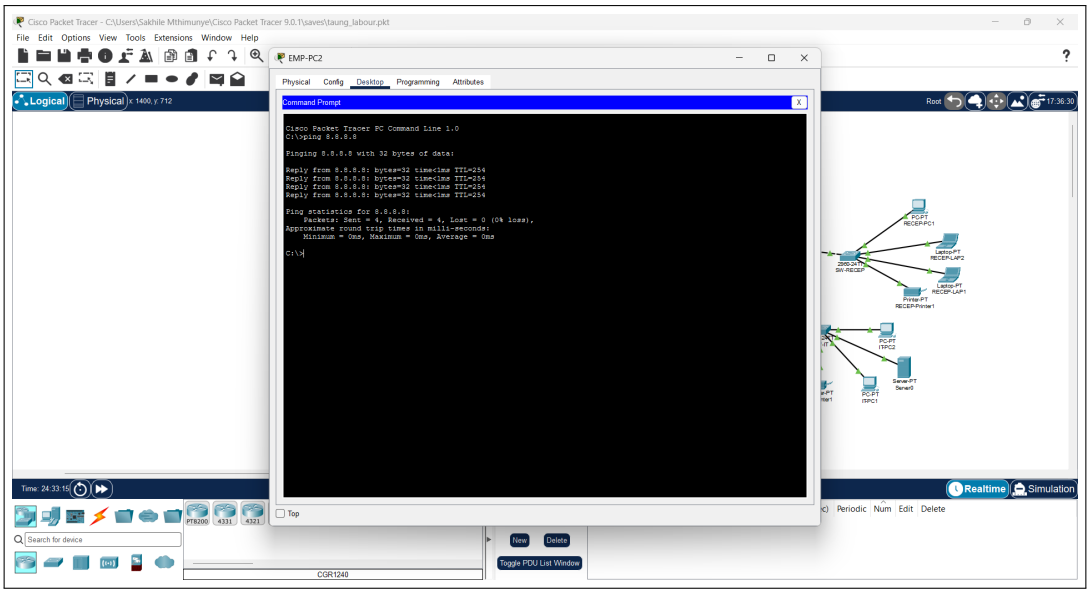


Figure 6: A PC in Employment Services pinging `8.8.8.8`: four replies, no loss.

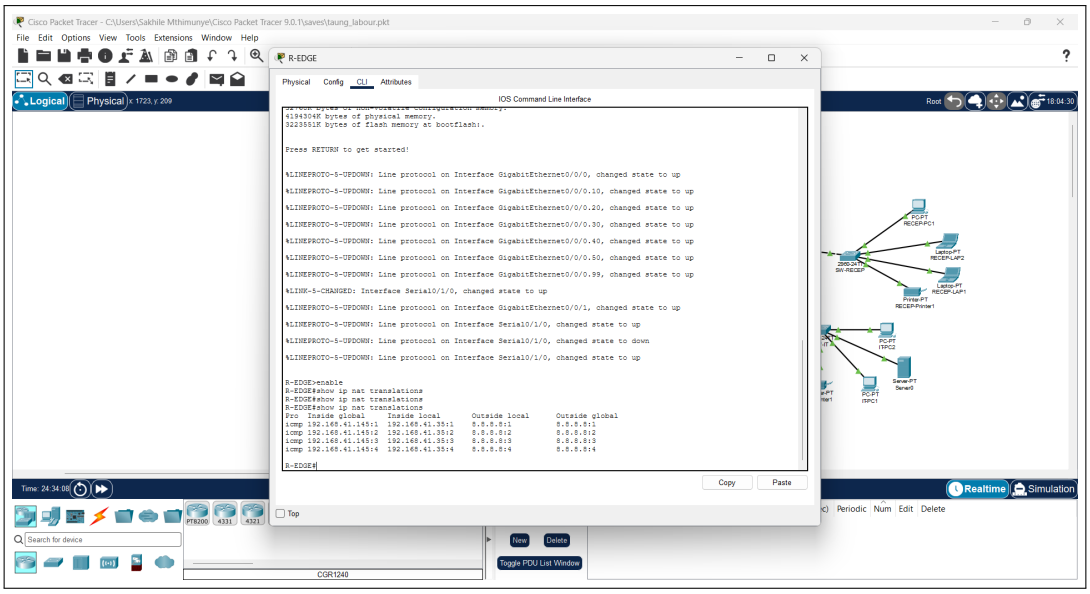


Figure 7: `show ip nat translations` on R-EDGE: private source `192.168.41.35` translated to `192.168.41.145`, the primary WAN interface address.

6 Assigned Challenge: Default Routing (Edge/ISP Path Design)

6.1 What was configured

```
ip route 0.0.0.0 0.0.0.0 192.168.41.146 1     ! primary: ISP-PRIMARY, AD 1
ip route 0.0.0.0 0.0.0.0 192.168.41.150 5     ! floating backup: ISP-BACKUP, AD 5
```

The first route is the preferred default route. The second is a *floating static route*: it points to the backup ISP but has a higher administrative distance (5 against 1), so the router keeps it out of the active routing table while the primary route exists.

6.2 Why this is appropriate

A router installs only the route with the lowest administrative distance. A static route normally disappears from the table when the interface carrying it goes down. Together this gives automatic failover and automatic recovery without a dynamic routing protocol, which matches an Intermediate challenge:

Primary link	Active default route	Gateway of last resort
Up	AD 1 via 192.168.41.146	192.168.41.146
Down	AD 5 via 192.168.41.150	192.168.41.150
Restored	AD 1 via 192.168.41.146	192.168.41.146

6.3 How the Finance constraint is met

All departments reach the internet through R-EDGE, so when the primary link fails, Finance keeps its internet access over the backup path together with every other department. The design does not give Finance a dedicated second path, which would need policy-based routing or a second edge device; the brief asks for a backup path for the Finance function, and this meets that requirement.

6.4 Verification

The failover was tested in four steps, and each step was captured.

Step 1: baseline. With both links healthy the gateway of last resort is the primary ISP.

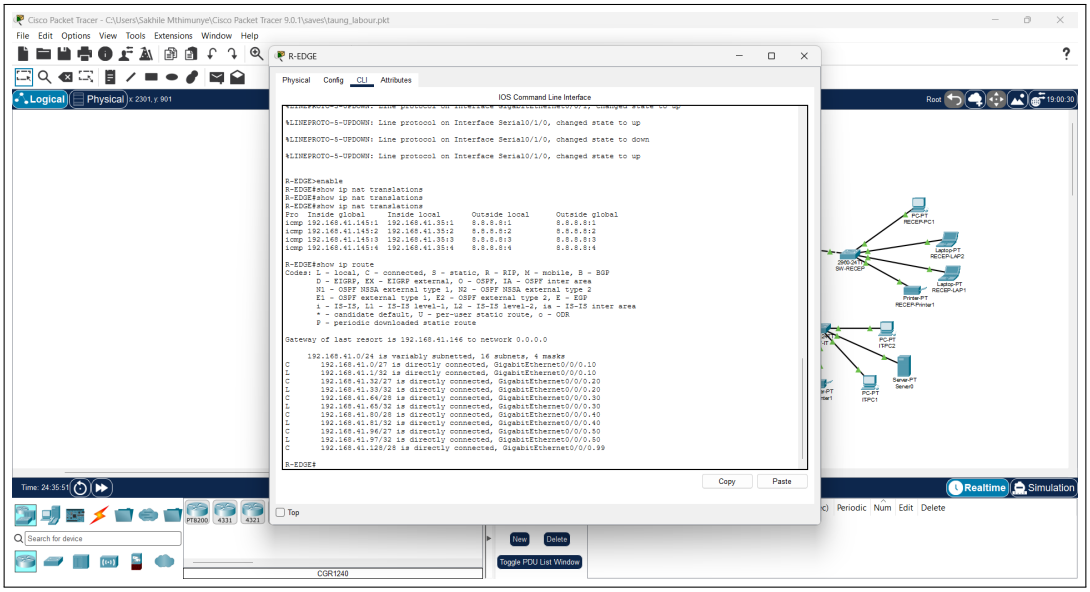


Figure 8: Before the outage: gateway of last resort `192.168.41.146` (16 subnets in the table).

Step 2: primary link down. `shutdown` was entered on `GigabitEthernet0/0/1`. The gateway of last resort changed to the backup ISP. The table also dropped from 16 to 14 subnets, because the connected route for the failed interface left the table.

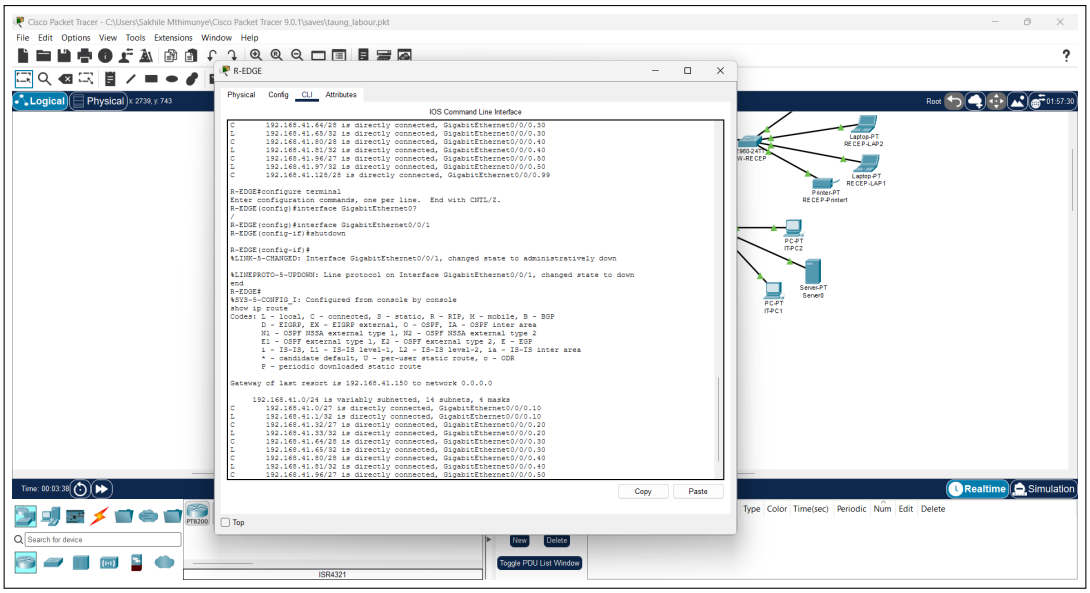


Figure 9: After `shutdown` on `Gi0/0/1`: gateway of last resort `192.168.41.150` (14 subnets).

Step 3: Finance during the outage. A Finance PC pinged `8.8.8.8` while the primary link was down and received four replies.

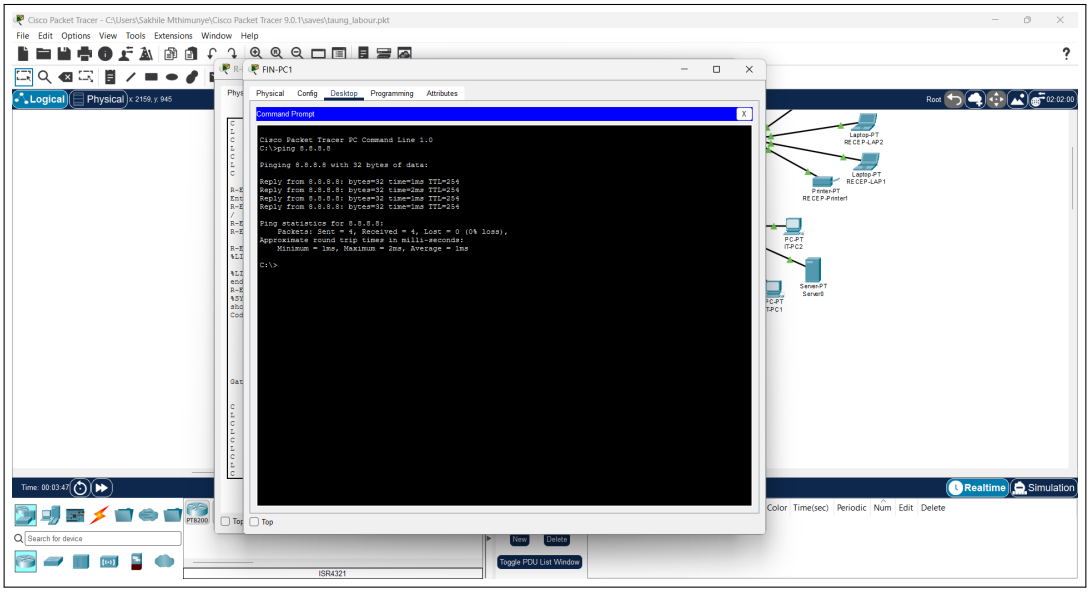


Figure 10: A Finance PC pinging `8.8.8.8` over the backup path: four replies, no loss.

Step 4: recovery. `no shutdown` restored the primary link. The router preferred it again and the table returned to 16 subnets.

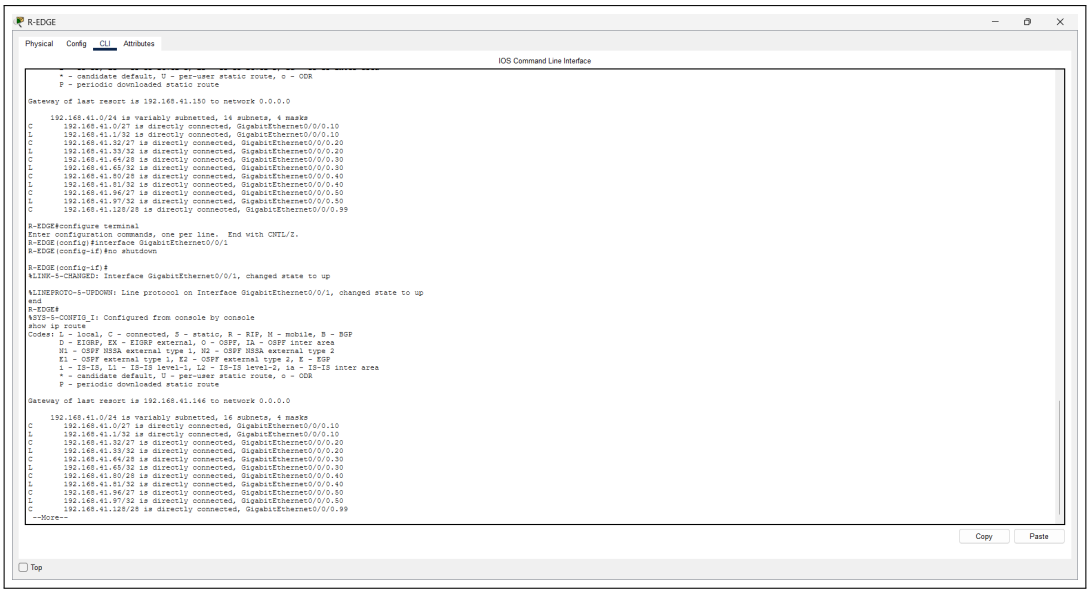


Figure 11: After `no shutdown` on `Gi0/0/1`: gateway of last resort back to `192.168.41.146` (16 subnets).

6.5 NAT during failover

Routing failover alone does not give working internet access, because the translated source address must also belong to the interface actually carrying the traffic. A first version with a single overload rule tied to the primary interface left no translation path when the primary was down (see Section 9). A second overload rule was added for the backup interface:

```
ip nat inside source list NAT_ACL interface GigabitEthernet0/0/1 overload
ip nat inside source list NAT_ACL interface Serial0/1/0 overload
```

7 Additional Feature: Reception Access Restriction

Reception is the public-facing network, so it should not be able to reach Finance or IT. An extended access list is applied inbound on R-EDGE's Reception sub-interface:

```
ip access-list extended RESTRICT_RECEPTION
 permit udp 192.168.41.96 0.0.0.31 host 192.168.41.85 eq domain
 permit tcp 192.168.41.96 0.0.0.31 host 192.168.41.85 eq domain
 deny ip 192.168.41.96 0.0.0.31 192.168.41.64 0.0.0.15
 deny ip 192.168.41.96 0.0.0.31 192.168.41.80 0.0.0.15
 permit ip any any
!
interface GigabitEthernet0/0/0.50
 ip access-group RESTRICT_RECEPTION in
```

Rule	Purpose
Permit DNS to Server0	Server0 is in the IT subnet, so DNS must be allowed before the deny rules or Reception could not resolve names.
Deny to Finance (`192.168.41.64/28`)	Protects the Finance VLAN from public-facing devices.
Deny to IT (`192.168.41.80/28`)	Protects servers and management devices.
Permit everything else	Reception keeps internet access and reaches Management and Employment Services.

The list is applied *inbound* on the Reception sub-interface so traffic is dropped as soon as it enters the router. Other departments remain open to each other, because the brief does not require broader segmentation.

Reception to Finance is blocked. The replies come from `192.168.41.97`, the router's Reception gateway, which is how the router reports traffic refused by the list.

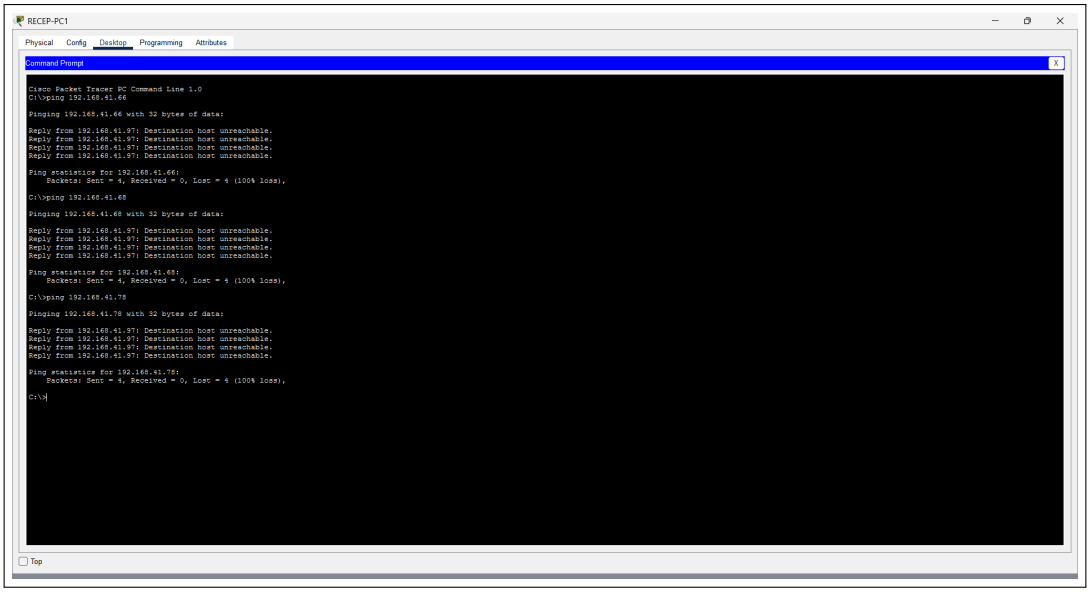


Figure 12: A Reception PC pinging Finance hosts: "destination host unreachable" from the Reception gateway, 100% loss.

Other departments are unaffected. A Management PC reaches the same Finance hosts. The first packet in each run times out while addresses are resolved, then the next three succeed.

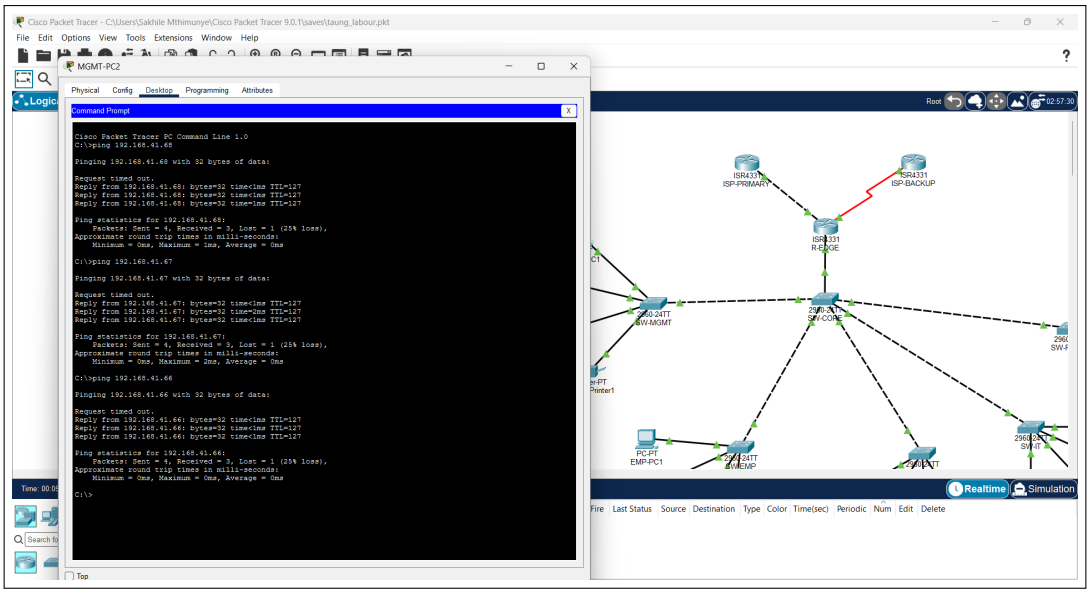


Figure 13: A Management PC pinging Finance hosts: three of four replies each time; the first request of each run times out.

DNS still works from Reception. Pinging the intranet name from a Reception PC shows the name resolved to `192.168.41.85`, which proves DNS traffic passed the list. The ICMP replies are then refused, as intended, because only DNS is permitted to that subnet.

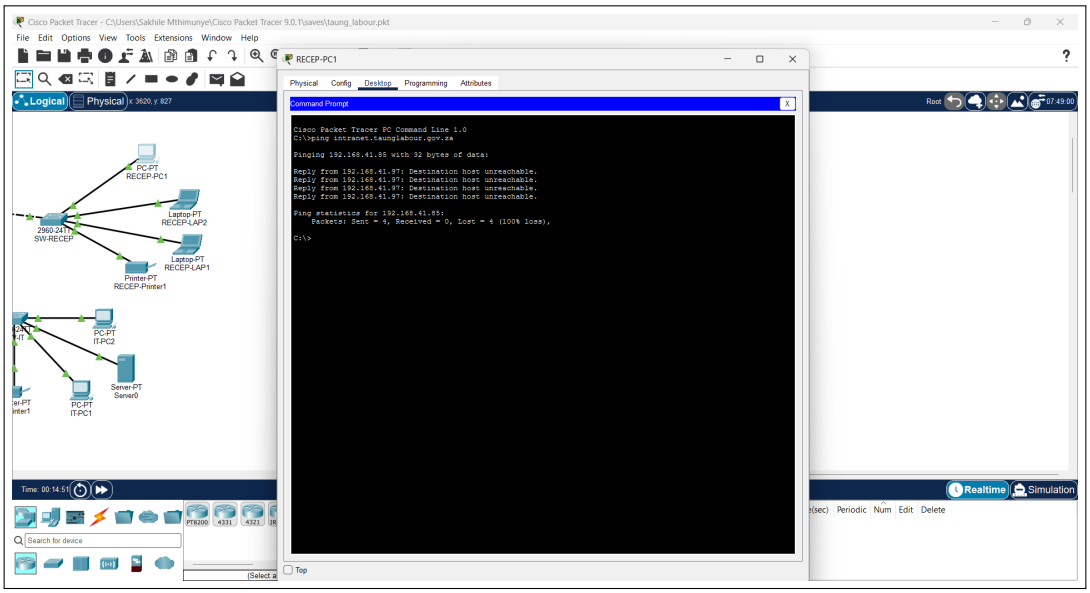


Figure 14: A Reception PC pinging the intranet name: the name resolves to 192.168.41.85, while ICMP to that subnet is refused.

8 Testing Summary

No.	Test	Expected result	Result	Evidence
1	VLANs on SW-CORE	VLANs 10, 20, 30, 40, 50, 99 active	Pass	Fig. 3
2	Trunk links	Six trunks, native VLAN 99, six VLANs allowed	Pass	Fig. 4
3	Routing table, primary healthy	Gateway `.146`, all VLAN subnets connected	Pass	Fig. 5
4	Internet reachability	PC pings `8.8.8.8`, four replies	Pass	Fig. 6
5	NAT	Translations through `192.168.41.145`	Pass	Fig. 7
6	Primary link down	Gateway changes to `.150`	Pass	Fig. 9
7	Finance during outage	Finance PC pings `8.8.8.8`, four replies	Pass	Fig. 10
8	Primary link restored	Gateway returns to `.146`	Pass	Fig. 11
9	Reception to Finance	Blocked	Pass	Fig. 12
10	Management to Finance	Allowed	Pass	Fig. 13
11	DNS from Reception	Name resolves to `192.168.41.85`	Pass	Fig. 14

9 Troubleshooting

Five problems were found and resolved. They are listed in the order they occurred, and the lesson from each is recorded in `reflection.md` in the repository.

No.	Symptom	Cause	Resolution
1	Every configuration line on a switch was rejected with "Invalid input"	Commands were entered at the user prompt (`Switch>`), not in configuration mode	Always enter `enable` then `configure terminal` before pasting configuration
2	`%Cannot create sub-interface` on the third router port	The added `NIM-ES2-4` module is a layer-2 switch module; it cannot hold an IP address or sub-interfaces, and `no switchport` did not convert it	Replaced it with a `NIM-2T` serial module. The backup ISP moved to `Serial0/1/0`; the LAN trunk stayed on the onboard `Gi0/0/0`, which supports sub-interfaces
3	Primary link "up" on one end but protocol "down", and no route installed	The IP address was configured on one port while the cable was on another (`show ip interface brief` on both devices exposed it)	Moved the configuration to the port that was actually cabled
4	`ping 8.8.8.8` failed from every PC	`8.8.8.8` did not exist anywhere in the simulation	Added a `8.8.8.8/32` loopback and a return route to each ISP device
5	`show ip nat translations` was empty and pings failed while the primary link was down	Traffic was correctly using the backup path, but the only NAT rule referred to the primary interface	Added a second overload rule for `Serial0/1/0` and brought the primary link back up for normal operation

Problems 2 and 5 are the most instructive. Both were hidden behind a configuration that looked correct: the first was a hardware limit that only appeared when sub-interfaces were created, and the second only appeared when failover was actually exercised.

10 GitHub Portfolio and Video

10.1 Portfolio

The individual GitHub portfolio is at https://github.com/SakhileGodfrery/Taung_Labour_Network. It contains the following:

```
README.md                project overview and milestone status
docs/                    client requirements, network design, addressing
                         plan, milestone 2 report, this technical report
diagrams/                physical and logical topology
configs/                 running configuration for every device
packet-tracer/           the final .pkt file
screenshots/             configuration and testing evidence
reflection.md            build notes and troubleshooting log
```

The commit history follows the order the work was done: repository scaffold, client requirements, topology, addressing plan, device configurations, the Reception restriction, baseline testing evidence, failover evidence, reflection, and the final files. Commits are small and describe one piece of work each.

10.2 Video demonstration

The 15–20 minute individual video, with an inset webcam view, accompanies this submission. It introduces the client and project, explains the requirements and design, demonstrates the Packet Tracer network and the assigned challenge, shows configuration and testing, and discusses design decisions, troubleshooting and a short reflection.

11 Reflection and Future Improvements

11.1 What was learned

The most useful lesson was the order in which to diagnose a fault: command mode first, then hardware capability, then physical cabling, then reachability, then behaviour under failure. Each of the five problems sat at a different layer, and assuming that the first thing that looked wrong was the cause would have wasted time. Testing failover deliberately, instead of assuming that a correct routing table meant working internet access, exposed the NAT limitation that a normal test would not have found.

11.2 Limitations

- **NAT selection:** two overload rules cover both outside interfaces, but choosing the rule per outgoing interface with route-maps is the more robust design for a production edge.
- **Finance backup:** the backup path protects the whole office, not Finance alone.
- **Single points of failure:** one edge router, one core switch and one DHCP/DNS server remain. The brief's constraint concerned the internet path only.
- **Access control:** only Reception is restricted. Other departments can reach each other, which the brief does not require changing.
- **Assumed headcounts:** the user numbers used for sizing are planning assumptions, since the brief gives none.

11.3 Possible improvements

Per-interface NAT with route-maps; policy-based routing to give Finance a dedicated path; a second DHCP/DNS server or DHCP on the router for resilience; and further access lists between departments if the client asks for stricter separation.

12 Conclusion

The network meets the client's requirements. Five department VLANs are routed through a single edge router with DHCP, DNS and NAT. The assigned Default Routing challenge was implemented with a primary route and a floating backup route, and it was tested through failure and recovery. The design constraint is met because Finance keeps internet access when the primary link fails. Change request CR5 is met because every subnet already holds its post-growth headcount, so no renumbering is needed. All of this is evidenced in the Packet Tracer file, the GitHub portfolio and the video demonstration.

A Device Configurations

Device	File	Role
R-EDGE	`configs/R-EDGE.txt`	Edge router
SW-CORE	`configs/SW-CORE.txt`	Core switch
SW-MGMT	`configs/SW-MGMT.txt`	Access switch, VLAN 10
SW-EMP	`configs/SW-EMP.txt`	Access switch, VLAN 20
SW-FIN	`configs/SW-FIN.txt`	Access switch, VLAN 30
SW-IT	`configs/SW-IT.txt`	Access switch, VLAN 40 (hosts Server0)
SW-RECEP	`configs/SW-RECEP.txt`	Access switch, VLAN 50

A.1 R-EDGE

```
! ============================================================
! R-EDGE - Edge router configuration
! Taung Labour Department Office | CMPG325-2026-095
! Paste into R-EDGE's CLI in global config mode (configure terminal)
!
! HARDWARE NOTE:
! - Gi0/0/0 (onboard) = LAN trunk to SW-CORE, carries the dot1Q
!   sub-interfaces for all 5 VLANs + native VLAN 99.
! - Gi0/0/1 (onboard) = Primary ISP link.
! - Serial0/1/0 (NIM-2T module) = Backup ISP link. A serial WIC/NIM
!   was used here instead of an Ethernet module because the
!   originally-tried NIM-ES2-4 is a Layer-2 switch module and
!   cannot take an IP address; NIM-2T's serial ports are always
!   routed, so this avoids that problem entirely.
! ============================================================

hostname R-EDGE
!
no ip domain-lookup
!
! -- LAN trunk to SW-CORE --
interface GigabitEthernet0/0/0
 description Trunk to SW-CORE
 no ip address
 no shutdown
!
interface GigabitEthernet0/0/0.10
 description VLAN 10 - Management
 encapsulation dot1Q 10
 ip address 192.168.41.1 255.255.255.224
 ip helper-address 192.168.41.85
 ip nat inside
!
interface GigabitEthernet0/0/0.20
 description VLAN 20 - Employment Services
 encapsulation dot1Q 20
 ip address 192.168.41.33 255.255.255.224
 ip helper-address 192.168.41.85
 ip nat inside
!
interface GigabitEthernet0/0/0.30
 description VLAN 30 - Finance
 encapsulation dot1Q 30
 ip address 192.168.41.65 255.255.255.240
 ip helper-address 192.168.41.85
 ip nat inside
!
interface GigabitEthernet0/0/0.40
 description VLAN 40 - IT / Servers
 encapsulation dot1Q 40
 ip address 192.168.41.81 255.255.255.240
 ip nat inside
```

```
!
interface GigabitEthernet0/0/0.50
 description VLAN 50 - Reception / Public
 encapsulation dot1Q 50
 ip address 192.168.41.97 255.255.255.224
 ip helper-address 192.168.41.85
 ip nat inside
 ip access-group RESTRICT_RECEPTION in
!
interface GigabitEthernet0/0/0.99
 description Native / switch management VLAN
 encapsulation dot1Q 99 native
 ip address 192.168.41.129 255.255.255.240
!
! -- WAN interfaces --
interface GigabitEthernet0/0/1
 description Link to Primary ISP
 ip address 192.168.41.145 255.255.255.252
 ip nat outside
 no shutdown
!
interface Serial0/1/0
 description Link to Backup ISP
 ip address 192.168.41.149 255.255.255.252
 ip nat outside
 no shutdown
! If "show controllers serial0/1/0" reports this end as DCE, uncomment:
! clock rate 64000
!
! -- Default routing (the assigned technical challenge) --
! Primary default route - administrative distance 1 (default), preferred.
ip route 0.0.0.0 0.0.0.0 192.168.41.146 1
! Floating static default route via Backup ISP - AD 5, only installed
! in the routing table if the primary route (AD 1) is withdrawn.
ip route 0.0.0.0 0.0.0.0 192.168.41.150 5
!
! -- Restrict Reception (public) from reaching Finance and IT --
ip access-list extended RESTRICT_RECEPTION
 permit udp 192.168.41.96 0.0.0.31 host 192.168.41.85 eq domain
 permit tcp 192.168.41.96 0.0.0.31 host 192.168.41.85 eq domain
 deny ip 192.168.41.96 0.0.0.31 192.168.41.64 0.0.0.15
 deny ip 192.168.41.96 0.0.0.31 192.168.41.80 0.0.0.15
 permit ip any any
!
! -- NAT (so private addressing can reach the internet) --
ip access-list standard NAT_ACL
 permit 192.168.41.0 0.0.0.255
!
ip nat inside source list NAT_ACL interface GigabitEthernet0/0/1 overload
ip nat inside source list NAT_ACL interface Serial0/1/0 overload
!
! NOTE: a second overload rule covers the backup interface so address
! translation still works while traffic is on the backup path. Selecting
! the rule per outgoing interface with route-maps is the more robust
! production design - mention it as a possible improvement.
!
end
write memory
```

A.2 SW-CORE

```
! ============================================================
! SW-CORE - Core switch configuration
! Taung Labour Department Office | CMPG325-2026-095
! Paste into SW-CORE's CLI in global config mode
! ============================================================

hostname SW-CORE
!
vlan 10
 name MANAGEMENT
```

```
vlan 20
 name EMPLOYMENT_SERVICES
vlan 30
 name FINANCE
vlan 40
 name IT_SERVERS
vlan 50
 name RECEPTION
vlan 99
 name NATIVE_MGMT
!
! -- Uplink to R-EDGE (port number depends on your Packet Tracer wiring) --
interface FastEthernet0/1
 description Trunk to R-EDGE
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,50,99
!
! -- Downlinks to the 5 access switches --
interface range FastEthernet0/2 - 6
 description Trunk to access switch
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,50,99
!
! -- Switch management SVI --
interface Vlan99
 ip address 192.168.41.130 255.255.255.240
 no shutdown
!
ip default-gateway 192.168.41.129
!
end
write memory
```

A.3 SW-FIN (example access switch)

The other four access switches use the same pattern, with their own VLAN number and management address.

```
! ============================================================
! SW-FIN - Finance department access switch
! This is the department the backup-ISP constraint protects -
! use it as the focal point of your failover demonstration.
! ============================================================
hostname SW-FIN
!
vlan 30
 name FINANCE
vlan 99
 name NATIVE_MGMT
!
interface FastEthernet0/1
 description Trunk to SW-CORE
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 30,99
!
interface range FastEthernet0/2 - 24
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
!
interface Vlan99
 ip address 192.168.41.133 255.255.255.240
 no shutdown
!
ip default-gateway 192.168.41.129
!
end
write memory
```

A.4 Simulated ISP devices

These two devices stand in for the internet and are not part of the client network.

```
! ISP-PRIMARY
hostname ISP-PRIMARY
interface GigabitEthernet0/0/0
 ip address 192.168.41.146 255.255.255.252
 no shutdown
interface Loopback0
 ip address 8.8.8.8 255.255.255.255
ip route 192.168.41.0 255.255.255.0 192.168.41.145

! ISP-BACKUP
hostname ISP-BACKUP
interface Serial0/1/0        ! serial port facing R-EDGE
 ip address 192.168.41.150 255.255.255.252
 no shutdown
interface Loopback0
 ip address 8.8.8.8 255.255.255.255
ip route 192.168.41.0 255.255.255.0 192.168.41.149
```